

Chapter 4 : Monte Carlo Method and Variance Reduction Methods

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Description of the method

We consider a numerical integration problem. We use the **Monte Carlo method** (MCM), which consists first of all in putting the quantity we are trying to calculate into the form of an expectation.

At the end of this step, we still need to calculate a quantity of the form $\mathbb{E}[X]$, where X is a random variable, it is necessary to be able to simulate random variables $X_1, \dots, X_N$ according to the law of X .

Mathematically, this means that we assume that we have the realization of a sequence of independent r.v. $X_1, \dots, X_N$, all following the law of X .

In computer terms, the simulation of an arbitrary distribution is reduced to that of a sequence of independent r.v. following a uniform distribution over the interval $[0, 1]$.

All that remains is to approximate $\mathbb{E}[X]$ by

$$\mathbb{E}[X] \simeq \frac{1}{N} (X_1 + \dots + X_N).$$

Description of the MC Method

We want to calculate an integral of the form

$$I = \int_{[0,1]^d} g(u_1, \dots, u_d) du_1 \cdots du_d.$$

We set $X = g(U_1, \dots, U_d)$ where $U_1, \dots, U_d$ are independent r.v. following the uniform distribution on the interval $[0, 1]$, we have

$$\mathbb{E}[X] = \mathbb{E}[g(U_1, \dots, U_d)] = \int_{[0,1]^d} g(u_1, \dots, u_d) du_1 \cdots du_d$$

by definition of the d -uplet law $(U_1, \dots, U_d)$. We've just carried out the first step: putting into expectation form.

Description of the Method MCM

For the simulation, let's assume that $(U_i)_{1 \leq i \leq d}$ is a sequence of independent r.v. following the uniform distribution on $[0, 1]$ and let's set $X_1 = g(U_1, \dots, U_d)$, $X_2 = g(U_{d+1}, \dots, U_{2d})$, and so on.

Then $(X_i)_{i \geq 1}$ is a sequence of random variables i.i.d following the law of X . and a good approximation of I is given by

$$I \approx \frac{1}{N} (X_1 + \dots + X_N).$$

Remarks :

- ① The method is extremely easy to program.
- ② the applicability of this method does not depend on the regularity of f , which can be simply integrable.

Description of the MC Method

Generalization

We often aim to evaluate an integral in $\mathbb{R}^d$, more generally of type

$$I = \int_{\mathbb{R}^d} g(x) f(x) dx$$

$$I = \int_{\mathbb{R}^d} g(x_1, \dots, x_d) f(x_1, \dots, x_d) dx_1 \cdots dx_d$$

with $f(x)$ positive and $\int f(x) dx = 1$.

Thus I can be expressed as $\mathbb{E}[g(X)]$ if X is a random variable in $\mathbb{R}^d$ with density f .

We can then approximate I by

$$\hat{I} \approx \frac{1}{N} \sum_{i=1}^N g(X_i).$$

Convergence and Limit of the MC Method

The problem is then to know how to simulate a random variable according to this law.

We'll start by answering two questions:

- When and why does this algorithm converge?
- What is the approximation precision?

Convergence and Limit of the MC Method

Strong Law of Large Numbers and Monte Carlo Method

Theorem

Let $(X_n)_{n \geq 1}$ a sequence of independent random variables all following the same law of the r.v. X . Assume that $\mathbb{E}[|X|] < +\infty$, then for almost any $\omega \in \Omega$

$$\mathbb{E}[X] = \lim_{n \rightarrow +\infty} \frac{1}{n} (X_1 + X_2 + \dots + X_n)(\omega).$$

Remark: The integrability assumption is essential for the MCM.

Convergence and Limit of the Method

Central Limit Theorem and Monte Carlo Method

To obtain confidence intervals for $\mathbb{E}[X]$, we use the central limit theorem, which gives an asymptotic of the error:

$$\epsilon_n = \mathbb{E}[X] - \frac{1}{n} (X_1 + X_2 + \dots + X_n) \sim \text{Gaussian centered}.$$

Theorem

Let $(X_n, n \geq 1)$ a sequence of i.i.d. random variables of integrable square ($\mathbb{E}[X^2] < \infty$). Let $m = \mathbb{E}[X]$ and $\sigma^2 = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

Then the sequence $\frac{\sqrt{n}}{\sigma} (m - \bar{X}_n) = \frac{\sqrt{n}}{\sigma} \epsilon_n$ converges in law to Z a random variable with a centered reduced Gaussian distribution.

Convergence and Limit of the Method

Central Limit Theorem and Monte Carlo Method

This means that

$$\lim_{n \rightarrow +\infty} \mathbb{E} \left[g \left(\frac{\sqrt{n}}{\sigma} \epsilon_n \right) \right] = \mathbb{E} [g(Z)] = \int_{-\infty}^{+\infty} g(x) \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx.$$

And for any pair of real numbers $a \leq b$ such that $a \leq Z \leq b$, we have

$$\lim_{n \rightarrow +\infty} \mathbb{P} \left(\frac{\sigma}{\sqrt{n}} a \leq \epsilon_n \leq \frac{\sigma}{\sqrt{n}} b \right) = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx.$$

Convergence and Limit of the Method

Central Limit Theorem and Monte Carlo Method

A table of the distribution function of a reduced centered Gaussian distribution shows that if Z is $\mathcal{N}(0, 1)$, $\mathbb{P}(|Z| \leq 1,96) = 0,95$.

We deduce that for n large enough,

$$\mathbb{P}\left(|\epsilon_n| \leq 1,96 \frac{\sigma}{\sqrt{n}}\right) \sim 0,95,$$

i.e. we have a 95% confidence interval for $\mathbb{E}[X]$ by taking

$$\left[\bar{X}_n - 1,96 \frac{\sigma}{\sqrt{n}}, \bar{X}_n + 1,96 \frac{\sigma}{\sqrt{n}}\right].$$

Variance Reduction Method

Introduction

We have seen that the convergence rate of the Monte Carlo method is of the order of $\frac{\sigma}{\sqrt{n}}$. To improve this method, there are numerous techniques called variance reduction methods, which aim to decrease the value of σ^2 .

The general idea is to provide an alternative representation in the form of an expectation for the quantity to be calculated:

$$\mathbb{E}[X] = \mathbb{E}[Y], \text{ such that } V(Y) \ll V(X).$$

By seeking to reduce variance, we will explore several methods.

Variance Reduction Methods

- ① Importance Sampling "Echantillonnage préférentiel"
- ② Control Variable "Variable de contrôle"
- ③ Antithetic Variables "Variables antithétiques"
- ④ Stratification Method "Méthode de stratification"
- ⑤ Conditional Expectation or Conditioning "Valeur moyenne ou conditionnement"

Importance Sampling

Suppose we want to calculate $\mathbb{E}[g(X)]$ where X is a random variable taking values in $\mathbb{R}^d$ with probability density f on $\mathbb{R}$, then

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) f(x) dx.$$

Let $\tilde{f}$ be the density of another random variable Y , such that $\tilde{f} > 0$ and $\int_{\mathbb{R}} \tilde{f}(x) dx = 1$. We can then express $\mathbb{E}[g(X)]$ as follows:

$$\begin{aligned} \mathbb{E}[g(X)] &= \int_{\mathbb{R}^d} \frac{g(x) f(x)}{\tilde{f}(x)} \tilde{f}(x) dx \\ &= \mathbb{E}\left[\frac{g(Y) f(Y)}{\tilde{f}(Y)}\right] \end{aligned}$$

where Y is an r.v. with the probability density $\tilde{f}$.

Importance Sampling

We therefore have another method of calculating $\mathbb{E}[g(X)]$, using n draws of $(Y, Y_1, \dots, Y_n)$ i.i.d. of probability density $\tilde{f}$. We approximate $\mathbb{E}[g(X)]$, for $(X_i)_{i=1}^n$ which are i.i.d. with the same distribution as X , as follows:

$$\mathbb{E}[g(X)] \approx \frac{1}{n} \sum_{i=1}^n g(x_i),$$

or equivalently

$$\begin{aligned} \mathbb{E}[g(X)] &\approx \frac{1}{n} \left(\frac{g(y_1) f(y_1)}{\tilde{f}(y_1)} + \dots + \frac{g(y_n) f(y_n)}{\tilde{f}(y_n)} \right) \\ &= \frac{1}{n} \sum_{i=1}^n \frac{g(y_i) f(y_i)}{\tilde{f}(y_i)}. \end{aligned}$$

Let us define $Z = \frac{g(Y)f(Y)}{\tilde{f}(Y)}$, if $\text{Var}(Z) \ll \text{Var}(g(X))$, then the algorithm is improved.

Importance Sampling

It is easy to calculate the variance of Z

$$\text{Var}(Z) = \mathbb{E}[Z^2] - (\mathbb{E}[Z])^2 = \int_{\mathbb{R}} \frac{g^2(x) f^2(x)}{\tilde{f}(x)} dx - (\mathbb{E}[g(X)])^2.$$

Suppose $g \geq 0$, we can choose $\tilde{f}$ as close as possible to $|g \times f|$ then normalize it (by dividing by $\int \tilde{f}(x) dx$) to obtain a density whose law is easily simulatable.

finally compare $\text{Var}(g(X))$ and $\text{Var}(g(Y) f(Y) / \tilde{f}(Y))$.

Remark:

The function $\tilde{f}$ is called Importance sampling "la fonction d'importance".

Control Variate

The idea is to find a random variable Y and a constant C such that $\mathbb{E}[X] = \mathbb{E}[Y] + C$ with $\text{Var}(Y) < \text{Var}(X)$.

We aim to express $\mathbb{E}[g(X)]$ in the form

$$\mathbb{E}[g(X) - h(X)] + \mathbb{E}[h(X)]$$

where $\mathbb{E}[h(X)]$ can be calculated explicitly, and:

$$\text{Var}(g(X) - h(X)) < \text{Var}(g(X)).$$

An MCM is then used to calculate $\mathbb{E}[g(X) - h(X)]$

$$\mathbb{E}[g(X)] \approx \frac{(g(X_1) - h(X_1)) + \dots + (g(X_n) - h(X_n))}{n} + \mathbb{E}[h(X)]$$

where $(X_i)_{i=1}^n$ are i.i.d random variables with the same distribution as X .

Antithetics Variables

Suppose we want to compute $I = \int_0^1 g(x) dx$, Since $x \mapsto 1 - x$ leaves the measure dx invariant on $[0, 1]$, we can write:

$$I = \int_0^1 \frac{g(x) + g(1-x)}{2} dx$$

Thus, we can calculate I as follows: draw n r.v. $(U_1)_{i=1}^n$ iid $\sim \mathcal{U}[0, 1]$ and approximate I by

$$\begin{aligned} I_{2n} &= \frac{1}{n} \left[\frac{1}{2} (g(u_1) + g(1-u_1)) + \cdots + \frac{1}{2} (g(u_n) + g(1-u_n)) \right] \\ &= \frac{1}{2n} \sum_{i=1}^n [g(u_i) + g(1-u_i)]. \end{aligned}$$

The quality of the approximation improves $\text{Var}(I_{2n}) < \text{Var}(I)$.

Stratification Method

Suppose we want to calculate $I = \mathbb{E}[g(X)] = \int g(x) f(x) dx$ where X follows the distribution f defined on a domain D . This leads to the idea of a basic Monte Carlo method:

$$I = \frac{g(X_1) + g(X_2) + \dots + g(X_n)}{n}$$

where $(X_i)_{i=1}^n$ are i.i.d. random variables with the same distribution as $X \in D$.

Stratification Method

A new idea involves the decomposition of I . Suppose D is divided into m subsets ($D = \bigcup_{i=1}^m D_i$), then we can write:

$$I = \sum_{i=1}^m p_i I_i$$

where,

$$I_i = \mathbb{E} [g(X) / X \in D_i] ,$$

with $p_i = \mathbb{P}(\{X \in D_i\})$ and $\sum_{i=1}^m p_i = 1$.

Stratification Method

For each $i = 1, \dots, m$, I can be approximated using the Monte Carlo method with n_i random samples $X_1^{(i)}, X_2^{(i)}, \dots, X_n^{(i)}$ which are i.i.d and follow the same distribution as $X \setminus X \in D_i$:

$$I_i \approx \hat{I}_i = \frac{g(X_1^{(i)}) + g(X_2^{(i)}) + \dots + g(X_n^{(i)})}{n_i}.$$

Thus, we have:

$$I \approx p_1 \hat{I}_1 + p_2 \hat{I}_2 + \dots + p_n \hat{I}_n.$$

Stratification Method

With the conditional variances σ_i , and using the convexity inequality for x^2 twice, $((\sum_{i=1}^m p_i a_i)^2 \leq \sum_{i=1}^m p_i a_i^2$ if $\sum_{i=1}^m p_i = 1$), to show that

$$\text{Var}(g(X)) \geq \sum_{i=1}^m p_i \text{Var}(g(X) / X \in D_i) \geq \left(\sum_{i=1}^m p_i \sigma_i \right)^2.$$

Conditional Expectation

The idea is to condition X by a r.v. Y which leaves the expectation unchanged but reduces the variance, thus

$$\mathbb{E}[X] = \mathbb{E}[X/Y], \text{ such that } \text{Var}(X/Y) \leq \text{Var}(X),$$

the difficulty lies in the explicit expression of the function

$$h(x) = \mathbb{E}[(X/Y = y)].$$

In the case where f is a probability density of the pair (X, Y) and we want to calculate:

$$I = \mathbb{E}[g(X, Y)] = \int g(x, y) f(x, y) dx dy.$$

Conditional Expectation

The first approximation idea is

$$I \approx \frac{1}{n} \sum_{i=1}^n g(x_i, y_i),$$

where $(X_i, Y_i)_{i=1}^n$ are i.i.d random variables with the same distribution as (X, Y) .

Since X and Y are independent, we then have:

$$\mathbb{E}[g(X, Y)] = \mathbb{E}[g(X, Y) / Y = y].$$

Conditional Expectation

Lemma

We can also show, by interpreting $h(x)$ as a conditional expectation, that

$$\text{Var}(h(X)) \leq \text{Var}(g(X, Y)).$$